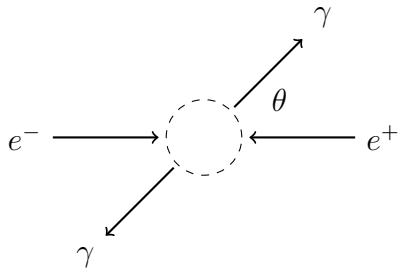
Annihilation

Annihilation is the process $e^- + e^+ \to \gamma + \gamma$.



Momentum vectors where $E = \sqrt{p^2 + m^2}$.

$$p_1 = \underset{e^- \longrightarrow}{\begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}} \qquad p_2 = \underset{\longleftarrow e^+}{\begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}} \qquad p_3 = \underset{\nearrow \gamma}{\begin{pmatrix} E \\ E\sin\theta\cos\phi \\ E\sin\theta\sin\phi \\ E\cos\theta \end{pmatrix}} \qquad p_4 = \underset{\gamma \swarrow}{\begin{pmatrix} E \\ -E\sin\theta\cos\phi \\ -E\sin\theta\sin\phi \\ -E\cos\theta \end{pmatrix}}$$

Spinors for p_1.

$$u_{11} = \frac{1}{\sqrt{E+m}} \underbrace{\begin{pmatrix} E+m \\ 0 \\ p \\ 0 \end{pmatrix}}_{\text{spin up}} \qquad u_{12} = \frac{1}{\sqrt{E+m}} \underbrace{\begin{pmatrix} 0 \\ E+m \\ 0 \\ -p \end{pmatrix}}_{\text{spin down}}$$

Spinors for p_2.

$$v_{21} = \frac{1}{\sqrt{E+m}} \underbrace{\begin{pmatrix} -p \\ 0 \\ E+m \\ 0 \end{pmatrix}}_{\text{spin up}} \qquad v_{22} = \frac{1}{\sqrt{E+m}} \underbrace{\begin{pmatrix} 0 \\ p \\ 0 \\ E+m \end{pmatrix}}_{\text{spin down}}$$

The scattering amplitude $\mathcal{M}$ is

$$\mathcal{M} = \mathcal{M}_1 + \mathcal{M}_2$$

where $\mathcal{M}_1$ is the t-channel amplitude

$$\mathcal{M}_1 = \frac{\bar{v}_{2b}(-ie\gamma^\mu)(\not{q}_1 + m)(-ie\gamma^\nu)u_{1a}}{t - m^2}$$

and $\mathcal{M}_2$ is the u-channel amplitude

$$\mathcal{M}_2 = \frac{\bar{v}_{2b}(-ie\gamma^\nu)(\not{q}_2 + m)(-ie\gamma^\mu)u_{1a}}{u - m^2}$$

Matrices $\not{q}_1$ and $\not{q}_2$ are momentum transfer.

$$\not{q}_1 = (p_1 - p_3)^\alpha g_{\alpha\beta}\gamma^\beta$$
$$\not{q}_2 = (p_1 - p_4)^\alpha g_{\alpha\beta}\gamma^\beta$$

Expected density $\langle|\mathcal{M}|^2\rangle$ is the average density.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}|\mathcal{M}|^2$$

Expand the summand. (Note indices μ and ν of $\mathcal{M}_2$ are interchanged relative to $\mathcal{M}_1$.)

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}\left(\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{1\mu\nu} + \mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{2\nu\mu} + \mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{1\mu\nu} + \mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{2\nu\mu}\right)$$

The summation results are

$$\sum_{ab}\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{1\mu\nu} = \frac{e^4}{(t-m^2)^2}\left(8tu - 24m^2t - 8m^2u - 8m^4\right)$$
$$\sum_{ab}\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{1\nu\mu} = \frac{e^4}{(t-m^2)(u-m^2)}\left(8m^2s - 32m^4\right)$$
$$\sum_{ab}\mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{2\nu\mu} = \frac{e^4}{(u-m^2)^2}\left(8tu - 8m^2t - 24m^2u - 8m^4\right)$$

where s, t, and u are Mandelstam variables

$$s = (p_1 + p_2)^2$$
$$t = (p_1 - p_3)^2$$
$$u = (p_1 - p_4)^2$$

For clarity define

$$f_{11} = 8tu - 24m^2t - 8m^2u - 8m^4$$
$$f_{12} = 8m^2s - 32m^4$$
$$f_{22} = 8tu - 8m^2t - 24m^2u - 8m^4$$

and write $\langle|\mathcal{M}|^2\rangle$ more compactly as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{f_{11}}{(t-m^2)^2} + \frac{2f_{12}}{(t-m^2)(u-m^2)} + \frac{f_{22}}{(u-m^2)^2}\right)$$

For $E \gg m$ a useful approximation is to set $m = 0$ and obtain

$$f_{11} = 8tu$$
$$f_{12} = 0$$
$$f_{22} = 8tu$$

Then $\langle|\mathcal{M}|^2\rangle$ simplifies as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{8tu}{t^2}+\frac{8tu}{u^2}\right) = 2e^4\left(\frac{u}{t}+\frac{t}{u}\right)$$

For $m = 0$ the Mandelstam variables are

$$\begin{aligned} s &= 4E^2 \\ t &= -2E^2(1-\cos\theta) \\ u &= -2E^2(1+\cos\theta) \end{aligned}$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4\left(\frac{1+\cos\theta}{1-\cos\theta}+\frac{1-\cos\theta}{1+\cos\theta}\right)$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4}\frac{\langle|\mathcal{M}|^2\rangle}{4s} = \frac{\alpha^2}{8E^2}\left(\frac{1+\cos\theta}{1-\cos\theta}+\frac{1-\cos\theta}{1+\cos\theta}\right)$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{8E^2}\left(\frac{1+\cos\theta}{1-\cos\theta}+\frac{1-\cos\theta}{1+\cos\theta}\right)(\hbar c)^2$$

Eigenmath script